Task 7: Introduction to Machine Learning (ML)

Objective

Understand the fundamentals of **Supervised and Unsupervised Learning**, including regression, classification, and clustering techniques using **scikit-learn**.

Implementation

- Split data into training and testing sets.
- Implement a **Linear Regression model** for prediction.
- Evaluate model performance using **R^2 score** and **Mean Squared Error (MSE)**.

Python Code

```
from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

import pandas as pd


# Load dataset

df = pd.read_csv("house_prices.csv")


# Define features and target

X = df[['sqft_living']]

y = df['price']


# Split dataset

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)


# Train model

model = LinearRegression()

model.fit(X_train, y_train)
```

```
# Evaluate model

y_pred = model.predict(X_test)

print("R² Score:", r2_score(y_test, y_pred))

print("MSE:", mean_squared_error(y_test, y_pred))
```

Client Project

Build a model to predict client data outcomes such as **house prices** or **sales growth**.